

Professional Summary

Electronics and Communication Engineering student with academic and project experience in FPGA-based digital design, MATLAB/Python simulation, Verilog HDL, and Xilinx Viado. Interested in electronics manufacturing, digital electronics, FPGA technology, testing, and quality. Eager to apply academic knowledge in an industrial environment while developing practical engineering skills.

Top Skills

- FPGA & Digital Design – Verilog HDL, behavioral simulation, synthesis, implementation, timing analysis and fixed-point arithmetic using Xilinx Vivado/XSim.
- Embedded & Electronics – Digital electronics, microcontroller fundamentals, STM32 basics, GPIO, ADC, PWM, timers and interrupts.
- Programming & Tools – C, C++, Python, MATLAB, Git, GitHub and VS Code.

Project Experience

FPGA Implementation of I-CDPK Codec

2026–Present

Final Year Project | Electronics & Communication Engineering

- Developing an FPGA implementation of Interleaved Chaotic Differential Peaks Keying (I-CDPK) modulation and demodulation for baseband digital communication.
- Modelled Lorenz chaotic X, Y and Z states and implemented peak detection, differential peak processing and interleaving through MATLAB/Python simulation and Verilog HDL.
- Performed fixed-point analysis using a 32-bit signed Q10.21 representation for FPGA-oriented implementation.
- Verified Verilog modules in Xilinx Vivado/XSim and completed synthesis, implementation and post-implementation timing verification for an Artix-7 target.
- Used Git and GitHub for version control, project organization and documentation.

Education

Priyadarshini College of Engineering, Hingna, Nagpur

2023–2027

Bachelor of Technology – Electronics and Communication Engineering | CGPA: 7.67/10

Nagpur, Maharashtra

Keshavnagar Madhyamik Vidyalaya & Junior Science College

2023

Higher Secondary | Percentage: 53.83%

Nagpur, Maharashtra

Keshavnagar Madhyamik Vidyalaya & Junior Science College

2021

Secondary Education | Percentage: 79.40%

Nagpur, Maharashtra

Certifications & Additional Information

- Cyber Security Course – [Introduction to cybersecurity Issued by Cisco/ Date issued: June 14th, 2024].
- Course 12: Motor Control Workshop using dsPIC® Digital Signal Controllers (DSC) Dual Core Devices [Issued by Microchip Technology/ Date issued: August 24th,2025]
- Tata – Cybersecurity Analyst job simulation [forage/ issued July 8th,2025]
- Languages: English, Hindi, Marathi.
- Add here: [Any additional embedded systems / STM32 / MATLAB / electronics / industrial certification].